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import cv2

# Open webcam
cap = cv2.VideoCapture(0)

# Read first frame
ret, frame1 = cap.read()
ret, frame2 = cap.read()

while cap.isOpened():

    # Difference between consecutive frames
    diff = cv2.absdiff(frame1, frame2)

    # Convert to grayscale
    gray = cv2.cvtColor(diff, cv2.COLOR_BGR2GRAY)

    # Blur to reduce noise
    blur = cv2.GaussianBlur(gray, (5, 5), 0)

    # Threshold
    _, thresh = cv2.threshold(blur, 20, 255, cv2.THRESH_BINARY)

    # Dilate to fill gaps
    dilated = cv2.dilate(thresh, None, iterations=3)

    # Find contours
    contours, _ = cv2.findContours(
        dilated, cv2.RETR_TREE, cv2.CHAIN_APPROX_SIMPLE
    )

    for contour in contours:
        if cv2.contourArea(contour) < 1000:
            continue

        x, y, w, h = cv2.boundingRect(contour)
```

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# Draw rectangle around detected motion
cv2.rectangle(frame1, (x, y), (x + w, y + h),
              (0, 255, 0), 2)

cv2.putText(frame1, "Motion Detected",
            (10, 30),
            cv2.FONT_HERSHEY_SIMPLEX,
            1, (0, 0, 255), 2)

cv2.imshow("Real-Time Surveillance", frame1)

# Update frames
frame1 = frame2
ret, frame2 = cap.read()

# Press 'q' to quit
if cv2.waitKey(30) & 0xFF == ord('q'):
    break

cap.release()
cv2.destroyAllWindows()
```

